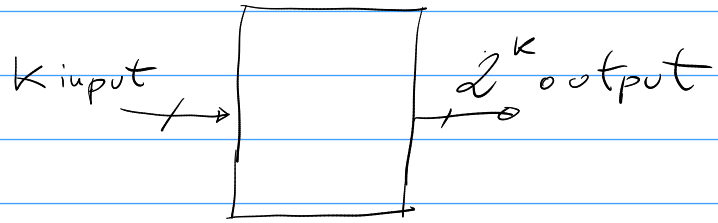
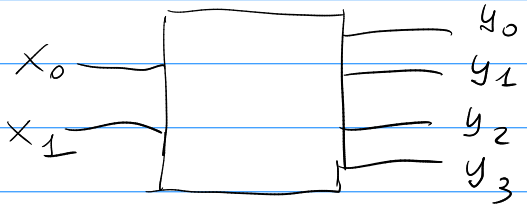
DECODER



DECODER A 2 BIT

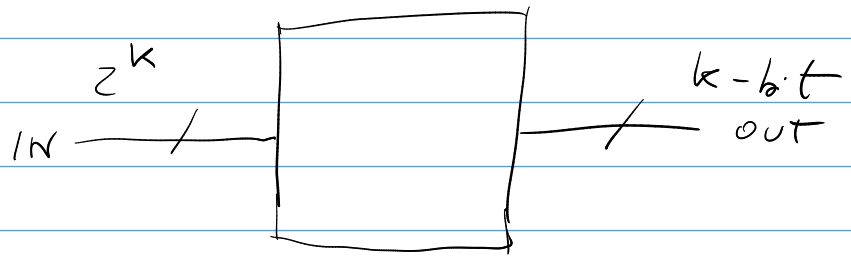


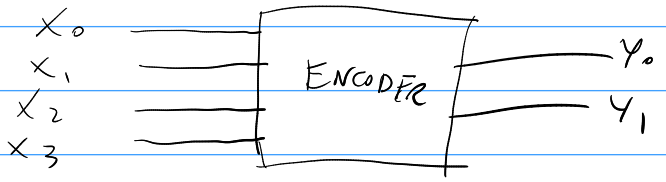
x_1	x_0	y_3	y_2	y_1	y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

$y_0 = \overline{x_1}\,\overline{x_0}$, $y_1 = \overline{x_1}\,x_0$, $y_2 = x_1\,\overline{x_0}$,

$y_3 = x_1\,x_0$

ENCODER A K-bit





x_3	x_2	x_1	x_0	y_1	y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

0 1 0 1 0

0 1 1 1 0

PRIORITA' ALL'INGRESSO DI ORDINE PIÙ BASSO

x_3	x_2	x_1	x_0	y_1	y_0
X	X	X	1	0	0
X	X	1	0	0	1
X	1	0	0	1	0
1	0	0	0	1	1

$$y_0 = x_1 \bar{x}_0 + x_3 \bar{x}_2 \bar{x}_1 \bar{x}_0$$

$$y_1 = x_2 \bar{x}_1 \bar{x}_0 + x_3 \bar{x}_2 \bar{x}_1 \bar{x}_0$$

x_3	x_2	x_1	x_0	y_1	y_0	VALID	$\overline{\text{VALID}}$
X	X	X	1	0	0	1	0
X	X	1	0	0	1	1	0
X	1	0	0	1	0	1	0
1	0	0	0	1	1	1	0
0	0	0	0	?	?	0	1

$$y_0 = x_1\overline{x_0} + x_3\overline{x_2}\,\overline{x_1}\,\overline{x_0}$$

$$y_1 = x_2\overline{x_1}\,\overline{x_0} + x_3\overline{x_2}\,\overline{x_1}\,\overline{x_0}$$

$$\overline{\text{VALID}} = \overline{x_3}\,\overline{x_2}\,\overline{x_1}\,\overline{x_0}$$